1.3 PROGRAMIMG BASICS

Programming : creative process

Techniques

» helpful when designing programming

» Applicable any PL, not particular to Java

Object ??

**ob·ject [abd?ikt | ?b-] n.

(오감(五感)으로 포착할 수 있는) 물건, 물체.

⊙ a tiny ~ 조그마한 물건.

a material thing that can be seen or touched.

Object-Oriented Programming: OOP

Real world : object

» Ex) people, automobile, buildings, trees, shoes, ships, sealing wax, cabbage, kings…

» 1) Perform

– Each of actions has some effect on some of other objects.

» 2) Have (colors,size, values..)

OOP : views a program as similarly consisting of objects that interact with one another by means of actions

Object-Oriented Programming: OOP

A design and programming technique

Some terminology:

- » [illegible] usually a (particular) person, place or thing (a noun)
- » [illegible] - an action performed by an object (a verb)
- » [illegible] - a category of similar objects (such as *automobiles)*

Objects have both data and methods

Objects of the same class have the same data elements and methods

Objects send and receive *messages* to invoke actions

Example of an (Object) Class

Class: Automobile

Data Items:

- » manufacturer's name
- » model name
- » year made
- » color
- » number of doors
- » size of engine
- » etc.

Methods:

- » Define data items (specify manufacturer's name, model, year, etc.)
- » Change a data item (color, engine, etc.)
- » Display data items
- » Calculate cost
- » etc.

Why OOP?

Save development time (and cost) by reusing code

» once an object class is created it can be used in other applications

Easier debugging

» classes can be tested independently

» reused objects have already been tested

Design Principles of OOP

Three main design principles of Object-Oriented Programming(OOP):

1) Encapsulation (??)

2) Polymorphism (??)

3) Inheritance (??)

Encapsulation

Encapsulation means to design, produce, and describe software so that it can be easily used without knowing

Also known as

An analogy:

When you drive a car, you don't have know the details of how many cylinders the engine has or how the gasoline and air are mixed and ignited.

Instead you only have to know the controls.(accelerator pedal, steering wheel…)

Polymorphism

Polymorphism—the same word or phrase can be mean different things in different

» Polymorphism :

Ex) Analogy: in English, **bank** can mean side of a river or a place to put money

Ex) analog/digital dual-mode cellular phone

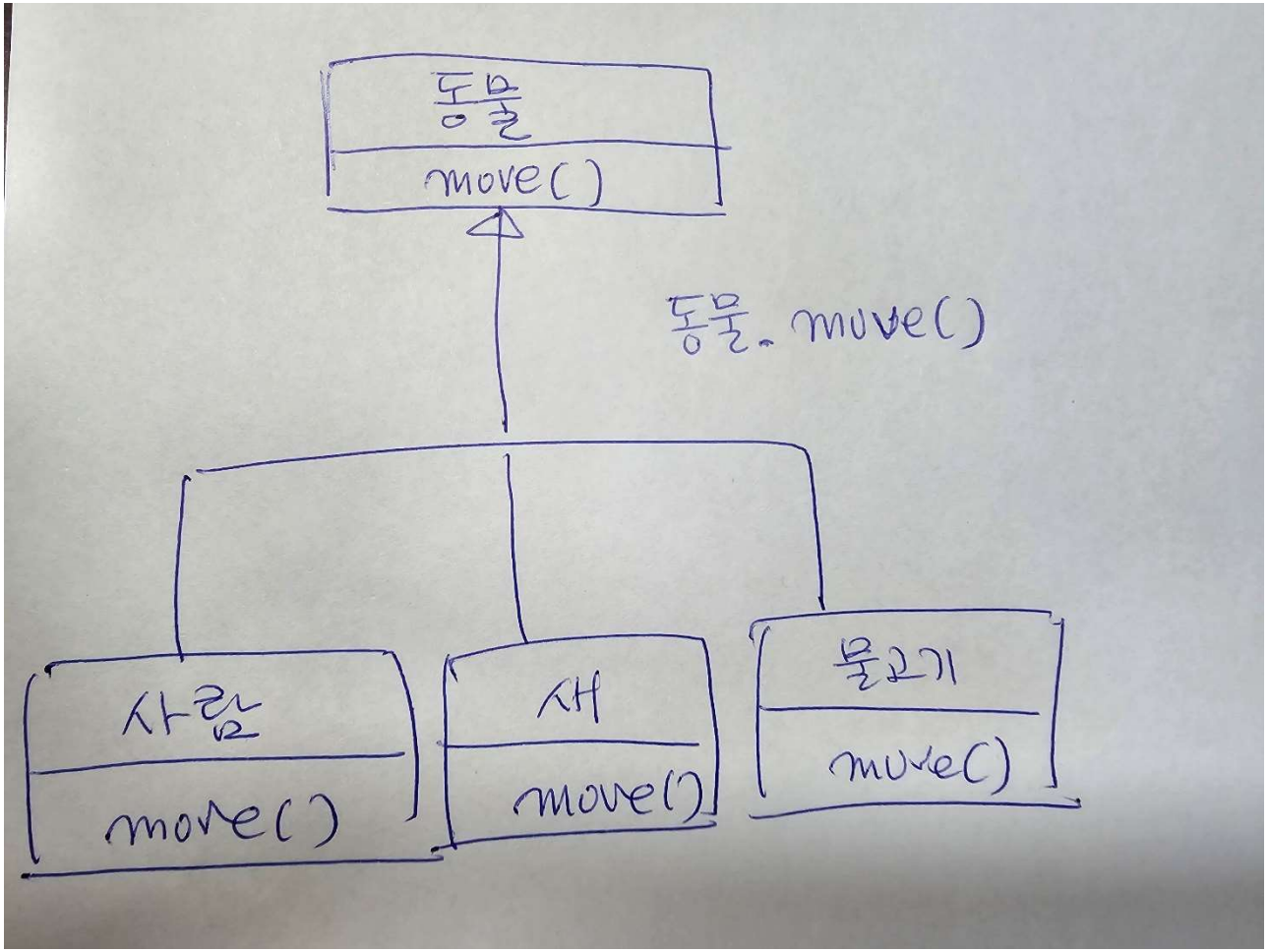
In programming (??)

Polymorphism

In programming, polymorphism means that one method name, used as an instruction, can cause different actions, depending on the kind of objects that perform the actions.

In Java, two or more classes could each have a method called **`output`**

- » Each **`output`** method would do the right thing for the class that it was in.
- » One **`output`** might display a number whereas a different one might display a name.


동물
move()
동물.move()
사람
move()
새
move()
물고기
move()

Inheritance

Inheritance—a way of organizing classes

Term comes from inheritance of traits like eye color, hair color, and so on.

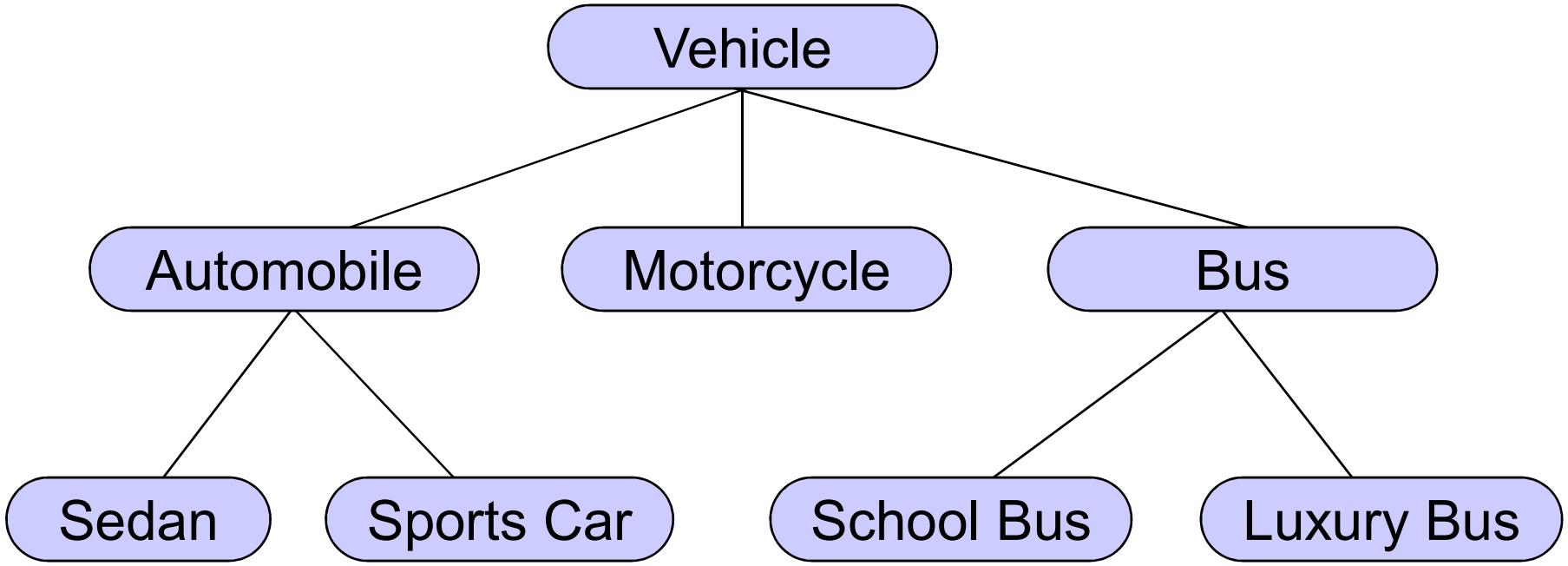
Classes with properties in common can be grouped so that <u>their common properties are only defined once.</u>

» <u>avoiding repeating the same set of programming instructions for each class.</u>

Display 1.4

» at each level, the classifications become more specialized.

Display 1.4 An Inheritance Hierarchy



What properties does each vehicle inherit from the types of vehicles above it in the diagram?

Algorithms

Algorithm - a set of instructions (steps) for solving a problem.

» must be precise

» must be complete

May be in a number of different formats

» natural language (such as English)

» a specific programming language

» a diagram, such as a flow chart

» Pseudo_code - a mix of natural and programming language

Example of an Algorithm

Algorithm that determines the total cost of a list of items:

1. Write the number 0 on the blackboard.

2. Do the following for each item on the list:
   --Add the cost of the item to the number on the blackboard.
   --Replace the old number on the board by this sum.

3. Announce that the answer is the number written on the board

Reusable Components

Advantages of using reusable components:

- saves time and money
- components that have been used before are often <u>better tested and more reliable</u> than new software

Make your classes reusable:

- encapsulation
- <u>general classes</u> have a better chance of being reused than <u>ad hoc classes</u>

Program Design Process

Design, then code

Design process

» define the problem clearly
» design objects your program needs
» develop <u>algorithms for the methods</u> of objects
» describe the algorithms, usually in pseudo_code
» write the code
» test the code
» fix any errors and retest

Testing and Debugging

Even with careful programming, your code could still contain errors and must be thoroughly tested.

Bug—a mistake in a program

Debugging—fixing mistakes in a program

Types of Errors

Syntax

Run-Time

Logic

Syntax

The set of grammar rules for a programming language is called the **syntax**.

The compiler checks your program to make sure it is a valid Java program.

If your program is not a valid Java program, then the compiler outputs a message indicating a **syntax error**.

Syntax Errors

- caught by compiler ("compiler-time error")
- automatically found, usually the easiest to fix
- cannot run code until all syntax errors are fixed
- error message may be misleading

Example:

- Misspelling a command, for example "rturn" instead of "return"

Run-Time Errors

An execution error (during run-time)

Not always so easy to fix

Error message may or may not be helpful

Not detected by the compiler.

Example:

Division by zero - if your program attempts to divide by zero it automatically terminates and prints an error message.

Logic Errors

Just because it compiles and runs without getting an error message does <u>not</u> mean the code is correct!

An error in the design (the algorithm) or its implementation

» code compiles without errors

» no run-time error messages

» but incorrect action or data occurs during execution

Generally the most difficult to find and fix

Need to be alert and test thoroughly

» think <u>about test cases</u> and predict results ***before*** executing the code

Logic Error Examples

Algorithm Error:

» **`averageOfFiveScores = SumOfScores/2`**
(should divide by 5)

Implementation Error:

» typed in wrong symbol in source code -
`sum = a - b;`
(should be **`sum = a + b;`**)

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